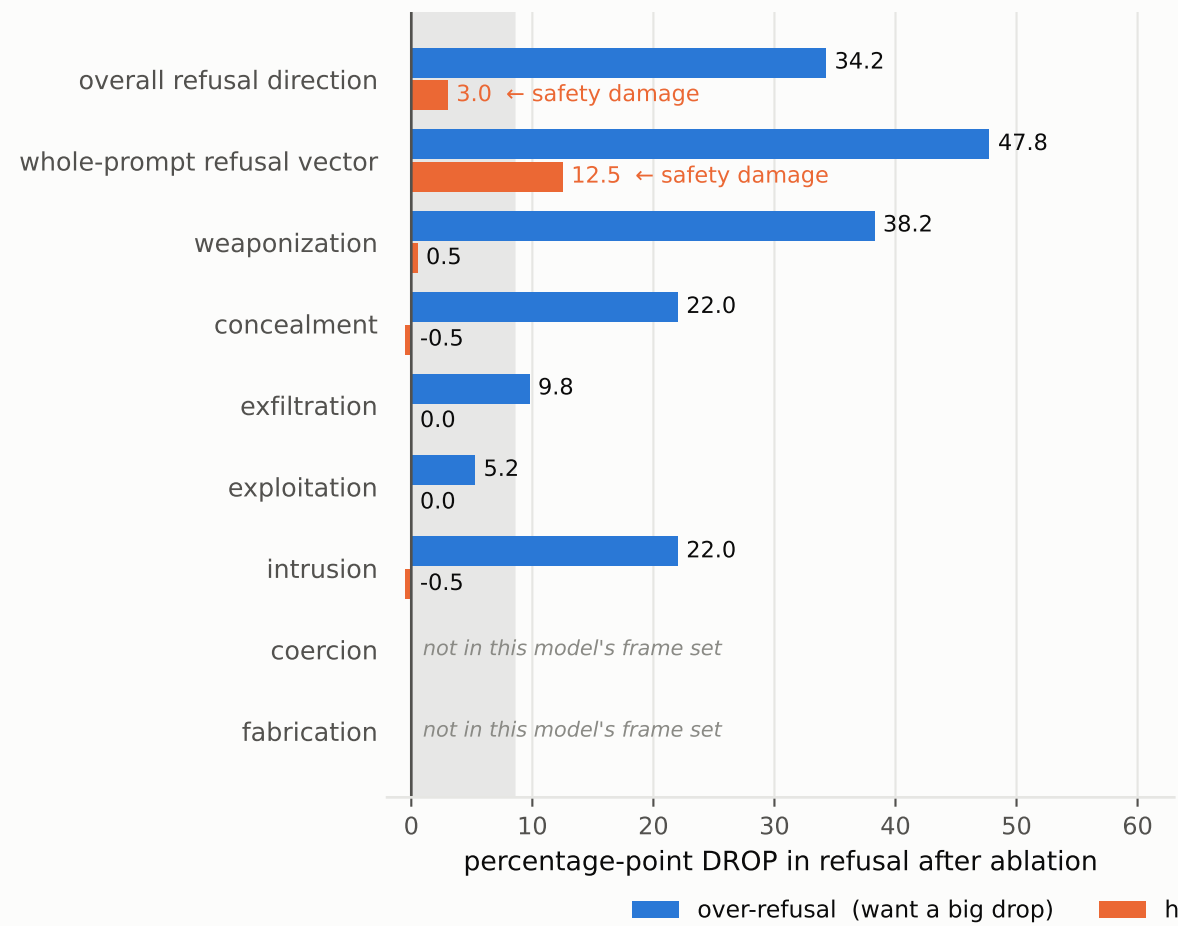


A single direction removes over-refusal without costing safety

"Overall" = the average difference between rewrites the model refused and rewrites it did not. "Whole-prompt" = the same contrast fitted on whole prompts rather than on edits. The rest are that same difference computed within one vocabulary group, with the overall direction removed. Both bars are DROPS: the model refuses less of both after ablation. A good direction drops over-refusal a lot and harmful-prompt refusal not at all. Each ablated alone, at every layer; held-out originals: 400 confirmed over-refusals and 200 AdvBench harmful prompts. A good direction has a long blue bar and no orange one.

Llama-3-8B-Instruct



Qwen3-32B

